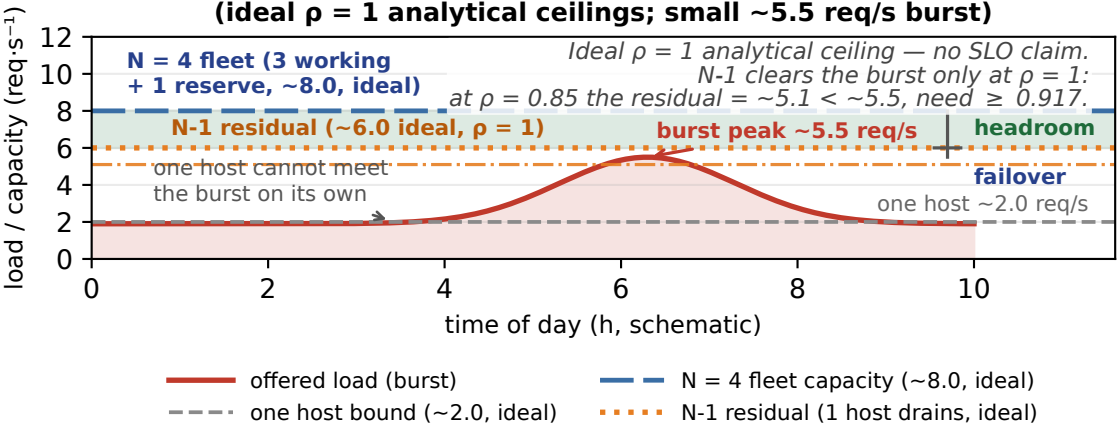
Sizing a fleet of one model
(ideal ρ = 1 analytical ceilings; small ~5.5 req/s burst)
load / capacity (req·s⁻¹)
time of day (h, schematic)
N = 4 fleet (3 working
+ 1 reserve, ~8.0, ideal)
Ideal ρ = 1 analytical ceiling — no SLO claim.
N-1 clears the burst only at ρ = 1:
at ρ = 0.85 the residual = ~5.1 < ~5.5, need ≥ 0.917.
N-1 residual (~6.0 ideal, ρ = 1)
burst peak ~5.5 req/s
headroom
one host cannot meet
the burst on its own
failover
one host ~2.0 req/s
offered load (burst)
N = 4 fleet capacity (~8.0, ideal)
one host bound (~2.0, ideal)
N-1 residual (1 host drains, ideal)